14 A car is driven over a series of equally spaced bumps separated by a distance of 5.0 m.

If the natural frequency of the car's suspension system is 2.0 Hz, what is the speed of the car for it to experience a large amplitude of vibration?

A $1.0\ m\ s^{-1}$

B $2.5\ m\ s^{-1}$

C $5.0\ m\ s^{-1}$

D $10\ m\ s^{-1}$